E01 - ESERCIZI CONVERSIONI TRA BASI (SOLUZIONI)

1

1500_{10} --> base 2

1500	0
750	0
375	1
187	1
93	1
46	0
23	1
11	1
5	1
2	0
1	1
0	

$1500_{10} = 10111011100_2$

765_{10} --> base 8

765	5
95	7
11	3
1	1
0	

$765_{10} = 1375_8$

112_{10} --> base 2

112	0
56	0
28	0
14	0
7	1
3	1
1	1
0	

$112_{10} = 1110000_2$

543_{10} --> base 8

543	7
67	3
8	0
1	1
0	

$543_{10} = 1037_8$

9999_{10} --> base 16

9999	15
624	0
39	7
2	2
0	

$9999_{10} = 270F_{16}$

22048_{10} --> base 16

22048	0
1378	2
86	6
5	5
0	

$22048_{10} = 5620_{16}$

455_{10} --> base 2

455	1
227	1
113	1
56	0
28	0
14	0
7	1
3	1
1	1
0	

$455_{10} = 111000111_2$

12350_{10} --> base 8

12350	6
1543	7
192	0
24	0
3	3
0	

$12350_{10} = 30076_8$

55600_{10} --> base 16

55600	0
3475	3
217	9
13	13
0	

$55600_{10} = D930_{16}$

2

$10001010_2 = 1*2^7 + 1*2^3 + 1*2^1 = 128 + 8 + 2 = 138_{10}$

$654_8 = 6*8^2 + 5*8^1 + 4*8^0 = 384 + 40 + 4 = 428_{10}$

$FF45_{16} = 15*16^3 + 15*16^2 + 4*16^1 + 5*16^0 = 61440 + 3840 + 64 + 5 = 65349_{10}$

$543_8 = 5*8^2 + 4*8^1 + 3*8^0 = 320 + 32 + 3 = 355_{10}$

$1110101011_2 = 1*2^9 + 1*2^8 + 1*2^7 + 1*2^5 + 1*2^3 + 1*2^1 + 1*2^0 = 512 + 256 + 128 + 32 + 8 + 2 + 1 = 939_{10}$

$2048_{16} = 2*16^3 + 4*16^1 + 8*16^0 = 8192 + 64 + 8 = 8264_{10}$

$BC12_{16} = 11*16^3 + 12*16^2 + 1*16^1 + 2*16^0 = 45056 + 3072 + 16 + 2 = 48146_{10}$

$7490_8 = 7*8^3 + 4*8^2 + 9*8^1 = 3584 + 256 + 72 = 3912_{10}$

$101101010_2 = 1*2^8 + 1*2^6 + 1*2^5 + 1*2^3 + 1*2^1 = 256 + 64 + 32 + 8 + 2 = 362_{10}$